

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- **Design and implement GenAI solutions** across multiple **LLM/multimodal** providers (OpenAI, Anthropic, Google), leveraging **ML infrastructure** and **evaluating tradeoffs between different techniques** and application domains via the **OpenAI Agents SDK** and **MCP** tool-use.
- Write and test production system code for a **multimodal** ingestion and transformation pipeline orchestrating multi-step generation workflows, in **Python** and **C++**, at **99%+ uptime**.
- Own a deployment-gated evaluation framework for **GenAI** model outputs: designing automated test and rollback gates, triaging regressions, and tracking issues to resolution, blocking **15%** of failed releases before production.
- Deployed a **RAG** retrieval and generation pipeline on **AWS SageMaker**, debugging and tuning for real-time decision support at **sub-200ms p99 latency**; collaborate with peers through design and code reviews.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Led the redesign of a production **ML infrastructure** retrieval architecture, improving recall **61% to 89%** on a 2K-query eval set via **hybrid dense-sparse retrieval**, re-ranking, and data processing pipeline changes.
- Built a voice-based **multimodal** conversational agent as part of the production agentic AI product surface, extending the platform beyond text-only interaction.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Systematically evaluated **CNN**, **ResNet**, **UNet**, and **Vision Transformer** architectures, optimizing augmentation, class balancing, and decision thresholds; improved computer-vision model accuracy **9.6 points** and cut false negatives **31%**.
- Designed and optimized **Python/C++ model-serving infrastructure** with async request handling, batching, health checks, and automated failure recovery, improving service availability to **99.9%** and cutting p99 inference latency **32%**.
- Profiled compute/memory bottlenecks and **optimized** batching, tensor movement, and request scheduling across **Python/C++** inference services, increasing throughput **2.4x** while cutting GPU memory usage **27%**.
- Architected **25+** distributed **C++/Java** services supporting **1M+ users** at **99.9% availability**, defining API, retry, timeout, and failure-isolation standards across the fleet that reduced production escalations **60%**.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- **Raft-consensus** distributed storage engine in **C++** handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **LangGraph** multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production **GenAI** evaluation platform with rubric-based scoring, hallucination detection, and regression alerting across model versions; **FastAPI** backend, **PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Java, TypeScript, SQL

Fundamentals: Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks

GenAI/ML: LLMs, Multimodal Pipelines, Prompt Engineering, RAG, ML Infrastructure, Model Evaluation/Optimization, PyTorch, TensorFlow

Computer Vision: CNN, ResNet, UNet, Vision Transformers, Image Segmentation

Web & Infra: HTML5, CSS, React, AWS (SageMaker, Lambda, EC2), GCP, Docker, Kubernetes, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer